



FABRIC MANUFACTURING II

Part: B

Course Code: FE 351-0723

Quantitative Aptitude

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TERMS

1. Rack: The unit of production in warp knitting is a 'rack'. A rack is 480 courses of knitting.
2. Run-in: The length of yarn consumed in knitting one rack is called a runner or a run in. Thus, the main factor controlling fabric quality is the rate of 'run in' of each warp. The precise figure for the run in will vary according to the lapping motion of the guide bar.
3. Quality: The length of cloth knitted during one rack measured in inches is called the quality.

PROBLEM 1

A 168", 28E tricot machine is operating at 800 courses per minute on a two bars machine. The warp is 70 denier nylon both in the front and back bars. The width of the fabric is 144 inches. The quality is eight inches per rack. The back bar run in is 64 inches and the front bar run in is 80 inches. Calculate

- i. Weight in lb per rack
- ii. Pounds per linear yards
- iii. Pounds per sq. yard
- iv. Racks per hour
- v. Wales per inch
- vi. Courses per inch
- vii. Pounds per hour
- viii. Run in ratio of the front bar to back bar

PROBLEM 1

Solution

(i) Weight in Pound (Lb) Per Rack: $\frac{\text{Bars} \times \text{Run in length} \times \text{Gauge} \times \text{Warp Width} \times \text{Denier}}{39.37 \times 9000 \times 453.6}$

$$= \frac{2 \times 72 \times 28 \times 168 \times 70}{39.37 \times 9000 \times 453.6}$$

$$= 0.295$$

(ii) Pounds per linear yards: $\frac{\text{lb/rack} \times 36}{\text{inch/rack}} = \frac{0.295 \times 36}{8} = 1.328$

(iii) Pounds per square yards: $\frac{\text{Pounds per Linear Yard} \times 36}{\text{Fabric Width (inch)}} = \frac{1.328 \times 36}{144} = 0.332$

PROBLEM 1

Solution

$$(iv) \text{ Racks Per Hour: } \frac{\text{Course Per Min} \times 60}{\text{Course Per Rack}} = \frac{800 \times 60}{480} = 100$$

$$(v) \text{ Wales Per Inch (WPI): } \frac{\text{Gauge} \times \text{Warp Width}}{\text{Fabric Width}} = \frac{28 \times 168}{144} = 33$$

$$(vi) \text{ Courses Per Inch (CPI): } \frac{\text{Courses Per Rack}}{\text{Inch Per Rack}} = \frac{480}{8} = 60$$

$$(vii) \text{ Pounds Per Hour: Pounds Per Rack} \times \text{Rack Per Hour} = 0.295 \times 100 = 29.5$$

$$(viii) \text{ Run in ratio of front bar to back bar: } \frac{\text{Run in of Front Bar}}{\text{Run in of Back Bar}} = \frac{80}{64} = 1.25$$

PROBLEM 2

Calculate the production per hour of a two bars machine running at 960 rpm at 90% efficiency. The top bar uses a 70 denier of 60" and the bottom bar a 20 Denier runner of 40", with the number of ends per bar being 4800.

Solution:

$$\text{Number of Racks Per Hr.} = \frac{RPM \times 60}{480} = \frac{960 \times 60}{480} = 120$$

$$\text{Production in lb/Hr} = \frac{\frac{Rack}{Hr} \times \text{Run in (inch)} \times \text{Total Ends} \times \text{Efficiency}}{39.37 \times 9000 \times 453.6}$$

$$\text{Top Bar Production in lb/Hr} = \frac{120 \times 70 \times 60 \times 4800 \times 0.90}{39.37 \times 9000 \times 453.6} = 13.54$$

$$\text{Bottom Bar Production in lb/Hr} = \frac{120 \times 20 \times 40 \times 4800 \times 0.90}{39.37 \times 9000 \times 453.6} = 2.580$$

So, Total Production in lb/Hr = 13.54 + 2.580 = 16.12 (Ans).

THANK
YOU !

